ParkNow: Low-fidelity Prototype

To follow the storyboard, along with the mockup and wireframe, please link on the link below:

https://www.figma.com/file/N7dfxspSHY5Cc0OGMyHbSc/Park-Now-Story-Board?type=whiteboard&node-id=0-1&t=6GwSTuyJF7ndz6Lh-0

Revised Requirements

1. As a user switching between different devices, I want the web application to be universally accessible across all web browsers so that I can effortlessly find parking as long as I have internet access.

2. As an occasional city visitor, I want the app to provide accurate directions to available parking spots so that I can navigate efficiently since I'm not familiar with the streets.

3. As an entry level income student, I want the app to allow me to filter parking spots based on proximity due to time work constraints and great affordability.
4. As a commuter using public transit, I want to know parking availability near transit stations so that I may easily integrate my various forms of transportation.

5. As a frequent city commuter and electric vehicle driver, I want the app to remember my preferences for parking spots so that I can quickly access suitable options every time I use it.
6. As an adult having a family and kids, I want the app to provide information on parking spot sizes to ensure they can accommodate my minivan sized vehicle.
7. As a new student driver, I want the app to recommend spacious parking spots so that I do not struggle to parallel park.
8. As a bicyclist, I want to identify secure parking options so that my bike is safe while I'm away.

Revised Personas

NAME
ELIZA WINDSOR

AGE
34

STATUS
Married & has children

PROFESSION
Housewife

RESIDENCE
Chicago Suburbs

VEHICLE
Minivan

BIO
Eliza is a suburban mother who values family time and convenience. Her visits to the city requires finding suitable parking spaces that can accommodate the size of her minivan. She values simplicity and intuitive design in technology, preferring straightforward solutions to enhance her driving experience.

Cherishing life's journey, I seek paths that offer space for my loved ones and simplicity in our adventures.

GOALS
To find an easy to navigate parking app that simplifies the process of locating large, family-friendly parking spaces.

To have access to features that help her stay focused and organized, accurately guiding her to correct parking spots.

PARKING CONSIDERATIONS
Convenient
Safe
Family-Friendly
Able to accommodate her minivan.

PERSONALITY TRAITS
Patience 85%
Family-Oriented 90%
Practicality 80%
Tech-Savvy 45%
Resourceful 75%
Detail-Oriented 60%

PARKING SEARCHING METHODS
Relies on street signs and visual cues while navigating the city. She avoids parking apps due to their complexity.

NAME
CARSON GOODWIN

AGE
23

STATUS
Single

PROFESSION
Software Engineer

RESIDENCE
Irving Park, Illinois

VEHICLE
Bicycle

BIO
Carson has graduated from school and is working full time at a company downtown. However, public transportation was his form of transportation. He realizes there have been a lot of delays and accidents in his commute that has caused him to be late at work. He has relied on his bicycle for commuting to work now, however as he is barely starting his career as a software engineer, he often struggles to find affordable parking especially in Downtown Chicago.

Every pedal stroke is a step towards a greener city.

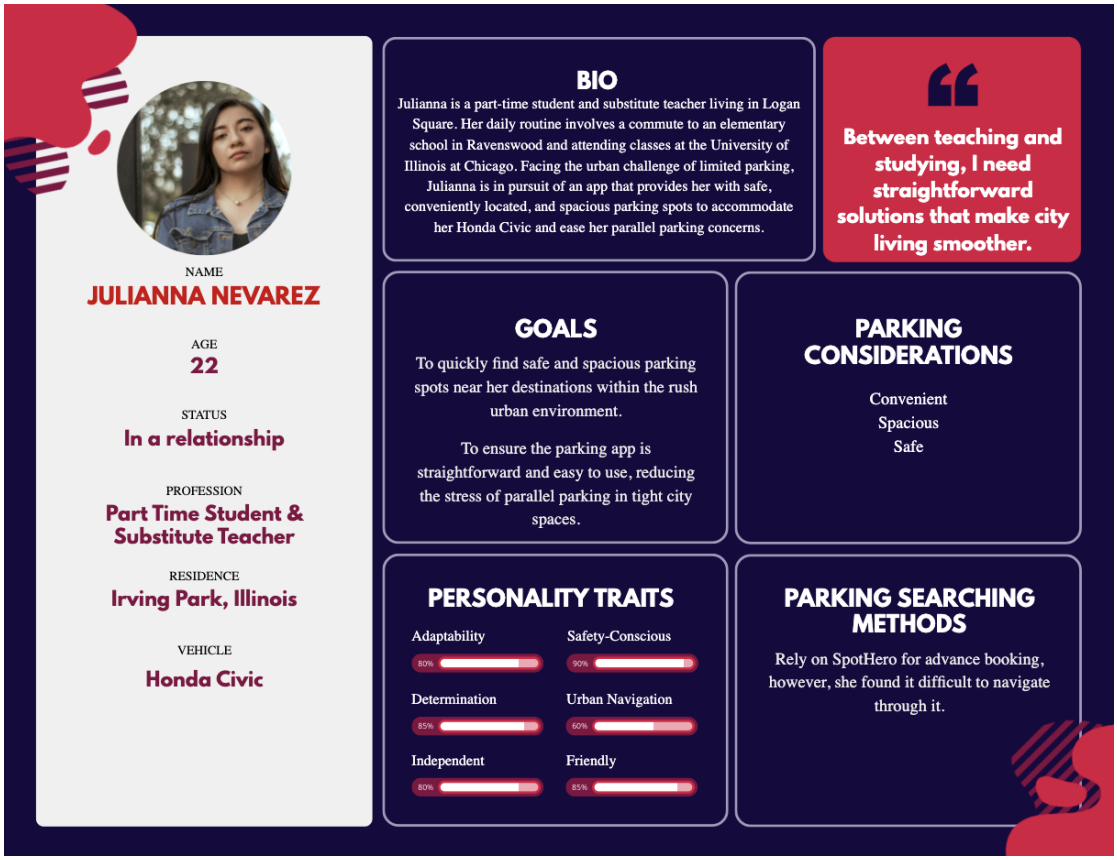
GOALS
To locate affordable and secure bike parking in proximity to his downtown workplace.

To receive timely notifications for parking deals and have the ability to reserve a spot in advance.

PARKING CONSIDERATIONS
Cost
Proximity to location
Safety

PERSONALITY TRAITS
Independence 75%
Cost-Conscious 85%
Eco-Friendly 95%
Tech-Savvy 70%
Health-Conscious 85%
Detail-Oriented 55%

PARKING SEARCHING METHODS
Rely on street signs and visual intuition to find parking spots while looking for parking near his job.



Revised Scenario

Scenario 1:
Miles Allison, an enthusiastic urban cyclist in Chicago, is dedicated to reducing his carbon footprint and embraces the freedom his bicycle provides for both his daily commute and social activities. However, this passion is tempered by the concern of finding secure parking in the city.

On a Friday evening, Miles pedals through downtown after work, heading to a cozy café to meet friends. The worry of bike security is ever-present; in Chicago, even a well-locked bike isn't immune to theft.

Before leaving work, Miles had opened the ParkNow website on his laptop. The community-driven platform, designed for urban cyclists, lists verified bike-friendly parking spaces. He had quickly checked the options and found a spot with positive reviews for security near the café. It was a simple process - find the location on ParkNow, note the spot, and then ride there.

As he nears the café, the dilemma of parking surfaces. The prime spots are already taken. He remembers the ParkNow recommended location and heads there instead of resorting to his

usual makeshift spot. The website had shown a picture of the space next to a coffee shop with a security camera overhead - perfect for keeping an eye on his bike.

Securing his bike at the spot he found through ParkNow, Miles meets his friends inside, his mind at ease for possibly the first time since he started his urban cycling. The evening passes with good food and better company, free from the usual frets about his bike's safety.

Returning to his bicycle, still securely parked, Miles is relieved. The ParkNow website had proven reliable. On his ride home, he appreciates the city's skyline, his thoughts not on the security of his bike, but on the success of the day and the simple yet effective solution ParkNow provided. It's a small victory, but for an urban cyclist, it's significant for a peace of mind.

Scenario 2:
Rosa, a dedicated system administrator, has been eagerly waiting for her promotion celebration with her friends. She spent Sunday evening celebrating her success and losing track of time and forgot to set an alarm for work the next morning.

With a sense of urgency, Rosa got ready and rushed out of her apartment hoping to make it to her office in time for her meeting at 9 AM. However, when Rosa reached the street and realized she had forgotten to reserve a parking space. Working downtown meant that the parking was always a challenge especially during peak hours. As Rosa reached close to her office building, she noticed that there were no available spots in the streets and everything was extremely expensive. Rosa usually uses SpotHero when she needs to reserve a spot and realizes that available parking spaces cost an outrageous amount of money for the amount of time she was going to be in this meeting.

Quickly, she accessed the ParkNow site on her mobile browser. The interface was straightforward, showing real-time availability of parking spots, with clear indications of price and location. With a couple of taps, she filtered the results for the best value within walking distance to her office. To her relief, ParkNow displayed several affordable options that had gone unnoticed by the harried commuters. She selected a spot that promised a reasonable rate for the duration of her meeting.

Rosa navigated directly to the chosen lot, found the ParkNow-listed space waiting for her, and secured her car. She was relieved and was able to attend her meeting on time.

Experience Design Concept
Our team has taken into consideration both the users' need for software accessibility across various platforms and the skill sets of our team members. As a result, we've made the decision for ParkNow to be a web application. We have noted that some users wish for their parking preferences set in previous sessions to be automatically applied without the need to reconfigure their filter options each time. To address this, we decided to implement a user account login system, storing their account information and preferences on Firebase.

In our selection of filter options, we prioritize those most relevant to our user base's parking search needs. This includes options such as parking cost and proximity. To ensure accurate proximity calculations, users will be prompted to input their final destination before available parking spots are displayed from their current location. Furthermore, recognizing the diverse range of vehicles our users may have, we will include a 'vehicle type' option. This allows for tailored parking suggestions, accommodating different vehicle types such as bicycles, which may require specialty parking spots, and larger vehicles that may not fit in standard parking spaces. We also have users that prioritize safety and family friendliness as their utmost priority, hence, “safeness” should be an option on the filters.

We discover that many users refuse to use parking apps in the market because of the complexity to navigate around the software. So, we will emphasize a simple and minimalistic user interface. To further promote safety, we willl include an eye catching bar on top of the web app displaying alert messages.

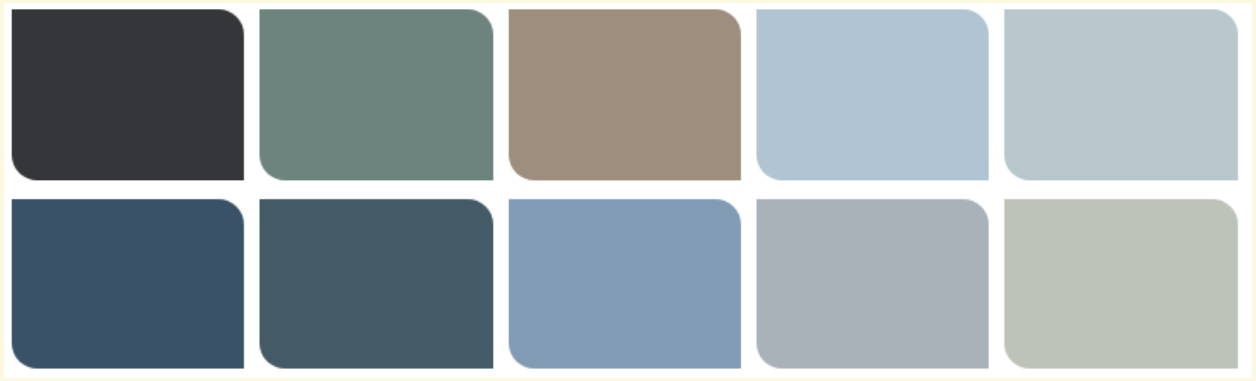
Many users expect a navigation feature that takes them from their current location to the chosen parking spot. However, this will not be addressed in this semester's scope of work since building or integrating a navigation software requires another complex layer of work. Parking reservations also won't be addressed within this semester's scope of work. Implementing them would demand extensive outreach initiatives, which is unrealistic and exceeds the capacity of our course project.

ParkNow starts with the transition from its logo to a screen that prompts the user to specify their purpose of visit. Users can choose between two options: "Find Parking" or "Post Parking".

Find Parking:
Upon selecting "Find Parking", the user is prompted to enter their final destination. The app obtains the user's current location through their phone settings with consent. After receiving the destination input, ParkNow displays a list of available parking spots on the embedded Google Maps interface and a list under the filtering section.

Post Parking:
Choosing "Post Parking" enables users to contribute by listing available parking spots. Users can input the parking attributes and conditions. Specifically the location can be input through addresses or current location. Once submitted, the posted parking spot becomes visible to other users. The time that the parking spot was still empty, when it was posted, should be highlighted because it could potentially be occupied after a while so it keeps parking searchers informed.

Color Scheme: warm neutral colors combined with cool toned colors + fonts: **Calistga**, Arial, Merriweather, and Lora



The “warm neutrals” palette consists of colors that are reminiscent of earthy tones. These colors are typically associated with stability, reliability, and approachability. The colors range from a light beige, which could be used for background or canvas areas to not overpower the user's visual senses, to darker shades of brown and green that could serve as excellent choices for text, icons, or elements that require emphasis and grounding. The warm rusty orange stands out as a potential call-to-action color or for highlights that need to attract the user's attention.

The "cool-toned" palette, on the other hand, leans towards a more professional and modern aesthetic. The deep blues convey trust and security, which are crucial for a parking app where users need to feel confident in the information provided. Lighter blues may offer a calming effect, suitable for backgrounds or less critical elements. The neutral grays can serve as secondary colors for text or interface elements, providing a visual break from the stronger blues and creating a balanced, harmonious interface.

For the typography, the use of fonts like Calistoga, Arial, Merriweather, and Lora suggests a mix of readability and elegance. Calistoga, with its bold and slightly whimsical character, could be useful for headings to draw attention. Arial, known for its clarity, would be excellent for body text and information that requires quick scanning. Merriweather, with its high legibility and slightly formal appearance, could be suitable for detailed information where readability is essential. Lora, with its well-balanced contemporary style, could be used interchangeably with Merriweather to maintain visual interest without sacrificing user experience.

This blend of warm, earth-inspired neutrals and cool, trust-inducing blues, combined with a curated selection of fonts, is used to create a visual language that is both welcoming and authoritative, aligning well with an app designed to provide a reliable and interactive service like parking solutions.

Storyboard:
The storyboard for the ParkNow web application offers a visual narrative of the user journey, highlighting key interactions within the platform. It is constructed to demonstrate two core functionalities: 'Find Parking' and 'Report Parking'.

The storyboard is punctuated with data visualizations representing user research, which informs the design and functionality of ParkNow. These include users' reliance on their phones for parking, primary transportation modes within urban areas, and the factors influencing their parking choices - location, cost, and security.

Each frame of the storyboard is carefully annotated to describe the user actions and system responses, ensuring clarity of the interaction flow. The use of real-world scenarios ensures that the storyboard remains grounded and relatable, providing a clear guide to the envisioned user experience. This visual program is not just a theoretical exercise; it is a practical tool that articulates the intended functionality of ParkNow, informed by user needs and behaviors, and crafted to provide an intuitive, helpful, and secure service for urban parking.